# Rohan Mishra

Data Analyst | Software Engineering Background

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LinkedIn | GitHub | Portfolio


## PROFILE

Data Analyst with a software engineering background in full-stack development (React, Node.js, TypeScript). Skilled in SQL, Python, and Power BI for end-to-end analysis — from database querying through statistical exploration to interactive dashboard reporting. Brings production-engineering discipline (clean code, reproducible pipelines, scalable systems) to data analysis work. Eager to apply analytical thinking and technical depth to solve real business problems with data.

## EDUCATION

**Bachelor of Computer Applications (BCA)**

Makhanlal Chaturvedi National University of Journalism and Communication, Bhopal — Satna, India | Expected Graduation: 2027

## SKILLS

**Data & Analytics:** SQL (SQLite), Python (pandas, matplotlib, seaborn), Power BI, Statistical Analysis, Data Cleaning & EDA, Data Visualization

**Engineering Background:** JavaScript, TypeScript, React.js, Next.js, Node.js, Express.js, PostgreSQL, MongoDB, Git, Docker, REST APIs

## PROJECTS

### OTT Streaming Platform Analytics — SQL, Python & Power BI | GitHub

*Tech Stack: SQLite, Python (pandas, matplotlib, seaborn), Power BI, SQL*

- Built an end-to-end analytics pipeline analyzing 2,500 streaming titles across Netflix, Prime Video, and Hotstar, integrating SQLite, Python, and Power BI into a single reproducible workflow.
- Wrote 13 SQL queries including window functions (RANK, ROW_NUMBER, LAG) and CTEs to surface platform benchmarking, year-over-year trends, and genre-level Pareto analysis.
- Conducted exploratory data analysis in Python, identifying a near-zero correlation (r = 0.18) between content rating and popularity, revealing that rating functions as a popularity ceiling rather than a direct driver.
- Designed and debugged an interactive Power BI dashboard with cross-filtering slicers, independently diagnosing and resolving a data-model relationship issue that broke filter propagation across pre-aggregated tables.
- Validated all SQL aggregations against independent pandas calculations, ensuring 100% consistency across both analysis layers.

### SyncBoard — Real-Time Collaborative Whiteboard | GitHub

*Tech Stack: Next.js, TypeScript, Socket.io, Express.js, Prisma, PostgreSQL, Zustand*

- Built a real-time collaborative whiteboard using Socket.io, achieving sub-100ms synchronization via WebSocket event broadcasting and optimized state reconciliation across multiple concurrent users.
- Implemented collaborative undo/redo using the Command Pattern, allowing users to selectively revert their own actions while preserving others' work, with support for 1,000-entry action history.
- Optimized network performance by reducing traffic by 40% through throttled emissions (60 FPS) and minimized database writes by 90% using debounced auto-save, maintaining a responsive UX for 5+ simultaneous users.
- Designed a scalable Zustand-based state architecture with room-based PostgreSQL persistence, shareable room URLs, and PNG/SVG export functionality, deployed on Vercel (frontend) and Railway (backend).

### Vortex — Geographically Distributed URL Shortener | GitHub

*Tech Stack: Next.js, TypeScript, Vercel Edge Functions, Prisma, PostgreSQL, Upstash Redis, QStash, Recharts, Tailwind CSS*

- Architected a high-performance URL shortener achieving sub-50ms global redirects by deploying redirect logic to Vercel Edge Functions across 100+ worldwide regions, eliminating central server latency.
- Implemented cache-first architecture using Upstash Redis for distributed caching, reducing database queries by 95% and achieving 10–20ms response times on cache hits with automatic cache invalidation on link deletion.
- Designed asynchronous analytics pipeline using QStash message queue, capturing device type, geographic location, browser data, and referrer information without impacting redirect performance, maintaining redirect times under 100ms.
- Built real-time analytics dashboard with auto-refreshing charts (SWR), search/filter functionality, and toast notifications, visualizing click trends, geographic distribution, and traffic sources using Recharts.
- Enforced privacy-conscious analytics by hashing IPs with SHA-256 before database storage and filtering automated traffic using the isbot library, achieving 95%+ analytics accuracy.